Engineering Enrollment and Wages: Full-Information Estimation of the Ryoo–Rosen Model

Max Cheong*
Guanghua School of Management, Peking University

October 24, 2026

Abstract

Students choose a field of study four years before they enter the labor market, and they are paid over a career, so enrollment depends both on how strongly students respond to wages and on how they form expectations of future wages. We estimate the Ryoo–Rosen model of the market for engineers by full-information maximum likelihood: the competitive equilibrium is solved as a linear-quadratic planning problem, cast in state-space form with measurement error, and taken to U.S. data for 1968–2010 with the Kalman filter. Enrollment is highly sensitive to persistent wage changes: a permanent 1 percent increase in the relative wage eventually raises the share of students entering engineering by about 6 percent. Yet most swings in engineering degrees come from changes in the attractiveness of the field, which follow a damped cycle of about 22 years, rather than from wages. Whether students look through temporary wage changes, as rational expectations imply, the data cannot tell: a cobweb model, in which they do not, fits about as well and better matches the comovement of wages and future degrees, but only with a nearly flat demand curve. The estimated demand elasticity therefore depends on how students form expectations.

Keywords: Maximum Likelihood Estimation, Rational Expectations, Kalman Filter, Linear-Quadratic Dynamic Programming, Human Capital Investment

*Thanks go to Professor Thomas J. Sargent for giving me advice on learning dynamic programming, the Kalman filter and maximum likelihood estimation, which is the toolkit of this paper. Data, code, and replication files are available at https://github.com/maxmaxmaxmaxmaxmax373/ryoo-rosen-mle.

1 Introduction

The supply of engineers in the United States has fluctuated sharply. Engineering accounted for 4.2 percent of all bachelor's degrees in 1976, 7.8 percent in 1985, and 4.7 percent in 2000.[1] The rise was preceded by a rise in the relative pay of engineers: between 1972 and 1982, the ratio of the median earnings of engineers to that of all college graduates rose from 1.33 to 1.50.[2] Such episodes are not new: talk of a "shortage" of engineers was widespread in the 1950s (Arrow & Capron 1959), and declining interest in engineering drew attention again around 1990 (Ryoo & Rosen 2004). They raise two questions: what drives the booms and busts in the market for engineers, and how far do students base their choice of major on the wages they expect to earn?

Freeman (1971, 1975, 1976) explained such cycles in the markets for scientists and engineers with a cobweb model: students choose a field on the basis of the wages prevailing when they enroll, while the supply they create reaches the market several years later. Siow (1984) and Zarkin (1985) studied occupational choice among lawyers and teachers under rational expectations, and Zarkin compared his model with a cobweb specification. Ryoo & Rosen (2004) developed a dynamic model of the engineering market in which firms' demand falls with the relative wage, students enroll on the basis of the earnings they expect after graduation, and graduates join the stock of engineers four years after they enroll. Estimating the model equation by equation on U.S. data for 1950–1990, they found a demand elasticity between 1.2 and 2.2 and an elasticity of enrollment with respect to career prospects between 2.5 and 4.5.

Our contribution is to estimate the Ryoo–Rosen model in a full-information framework, using CPS and NSF data for 1968–2010. We show that the competitive equilibrium solves a linear-quadratic planning problem, solve for its law of motion, write it in state-space form together with measurement error, and estimate all parameters jointly by maximum likelihood with the Kalman filter. The framework imposes the restrictions that rational expectations place across the demand, enrollment, and stock equations; it treats expected

[1] Engineering bachelor's degrees divided by all bachelor's degrees, from National Science Foundation, National Center for Science and Engineering Statistics (2013), Table 5.

[2] Author's calculations from IPUMS CPS ASEC microdata (Flood et al. 2025); Section 3 gives the definitions.

lifetime earnings, which are not observed, as a function of the model's state rather than measuring them from earnings profiles or instrumenting them; and it accounts for noise in the survey data. Within it we let demand and preference shocks follow unrestricted second-order autoregressions, which allow for cycles, and we construct confidence intervals from the profile likelihood. Because a cobweb model fits into the same likelihood, we can also compare the two forms of expectations directly.

We report three findings. First, enrollment is highly sensitive to persistent changes in wages: a permanent 1 percent increase in the relative wage eventually raises the share of students choosing engineering by about 6 percent under rational expectations and by about 5 percent under the cobweb, and the data rule out an insensitive response. Second, the large swings in the degree share come mainly from changes in the attractiveness of engineering itself, which follow a damped cycle of about 22 years, while the observed stock of engineers is dominated by measurement error. Third, the data cannot tell whether students look through temporary wage changes. Under rational expectations they do: a demand-driven wage increase, which decays by half in about four years, raises enrollment only about one-sixth as much as a permanent one, and for this reason the model understates the comovement of wages and future degrees. A cobweb model, in which students take the current wage as permanent, matches this comovement and fits about as well, but only with a nearly flat demand curve and an almost deterministic demand cycle. The answer matters for the demand side: the demand elasticity is about 1.6 under rational expectations, in line with Ryoo & Rosen (2004), and almost four times as large under the cobweb.

The method belongs to the literature that estimates linear rational-expectations models by full-information maximum likelihood. Hansen & Sargent (1980) showed how to formulate and estimate such models and stressed the restrictions that rational expectations impose across equations; Sargent (1989) showed how to add measurement error and evaluate the likelihood with the Kalman filter; and Hansen & Sargent (2013) developed the recursive framework systematically. Applications include dynamic labor demand (Sargent 1978), time to build (Altug 1989), overtime and the business cycle (Hall 1996), and hybrid DSGE–VAR models (Ireland 2004). We apply this approach to a labor market with a training lag, in which the stock of workers is a slowly moving state and the expected returns to entry

are not observed. The cost of joint estimation is that misspecification in any one equation affects all the estimates.

Section 2 presents the model and Section 3 the data. Section 4 reports the estimates and the fit of the model, Section 5 studies the dynamics, Section 6 compares rational and cobweb expectations, and Section 7 concludes.

2 The Model

We follow Ryoo & Rosen (2004). Time is discrete, and a period is one year. All variables are log ratios of engineers to all college graduates, measured as deviations from trend: w_t is the relative wage, N_t the relative stock of engineers, and n_t the relative share of students beginning college in year t who choose engineering, which we call the enrollment share. Training takes four years, so the degree share is $\pi_t = n_{t-4}$. A representative firm demands engineering services, and a representative household chooses enrollment and owns the stock of engineers. Both take the wage as given.

Two shocks drive the model. A demand shock θ_t shifts the marginal product of engineering services, and a preference shock ψ_t shifts the attractiveness of engineering to students. Each follows a stationary second-order autoregression,

$$\theta_{t+1} = a_1^\theta\,\theta_t + a_2^\theta\,\theta_{t-1} + \sigma_\theta\,\varepsilon_{t+1}^\theta, \tag{1}$$

$$\psi_{t+1} = a_1^\psi\,\psi_t + a_2^\psi\,\psi_{t-1} + \sigma_\psi\,\varepsilon_{t+1}^\psi, \tag{2}$$

where ε_t^θ and ε_t^ψ are independent i.i.d. standard normal innovations. We do not restrict the roots to be real. When $a_1^2 + 4a_2 < 0$, the roots are a complex pair $re^{\pm i\omega}$, and the impulse response is an oscillation with a period of $2\pi/\omega$ years that decays at rate $r = \sqrt{-a_2}$ (Appendix A.3). The discount factor is fixed at $\beta = 0.96$.

2.1 Firms, Households, and the Stock of Engineers

The firm. The firm chooses its use of engineering services $\{N_t^d\}$ to maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t \Big[(f_1 + \theta_t)\, N_t^d - \tfrac{f_2}{2}(N_t^d)^2 - w_t N_t^d \Big], \qquad f_2 > 0. \tag{3}$$

The problem is static, and the first-order condition equates the marginal product of engineering services with the wage. The constant f_1 affects only the mean wage and drops out in deviations from trend, which gives the inverse demand curve

$$w_t = -f_2 N_t^d + \theta_t. \tag{4}$$

Because the variables are in logs, $1/f_2$ is the elasticity of demand.

The stock of engineers. If a fraction $1-\delta$ of engineers leaves the profession each year and students graduate after four years, the number of engineers evolves as $L_{t+1} = \delta L_t + S_{t-3}$, where S_t is the number of students who begin engineering in year t. Log-linearizing around a balanced growth path with growth rate g, and subtracting the same equation for all college graduates, gives (Appendix A.1)

$$N_{t+1} = \lambda N_t + (1-\lambda)\, n_{t-3}, \qquad \lambda = \frac{\delta}{1+g}. \tag{5}$$

The parameter λ is the share of incumbents in next year's stock. Because each graduating cohort is small relative to the stock, the stock adjusts slowly.

The household. The household chooses the enrollment share $\{n_t\}$ to maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t \Big[w_t N_t^h - \tfrac{d}{2}\, n_t^2 + d\, \psi_t\, n_t \Big] \quad \text{subject to} \quad N_{t+1}^h = \lambda N_t^h + (1-\lambda)\, n_{t-3}. \tag{6}$$

The first term is the wage income of the stock of engineers that the household owns. The marginal cost of enrollment, $d\, n_t$, rises with the enrollment share because potential students differ in ability and in taste for engineering, and the preference shock adds a nonpecuniary return $d\,\psi_t$ per entrant, so the net marginal cost of enrollment is $d\,(n_t - \psi_t)$. A cohort that enrolls in year t raises the stock in year $t+4+j$ by $(1-\lambda)\lambda^j$, and the first-order condition for n_t is (Appendix A.2)

$$d\,(n_t - \psi_t) = (1-\lambda)\, E_t \sum_{j=0}^{\infty} \beta^{4+j} \lambda^j\, w_{t+4+j}. \tag{7}$$

Rearranging gives the enrollment equation

$$n_t = \frac{1-\lambda}{d}\, V_t + \psi_t, \qquad V_t = \beta^4\, E_t \sum_{j=0}^{\infty} (\beta\lambda)^j\, w_{t+4+j}. \tag{8}$$

V_t is the career value of engineering: the expected present value of the relative wages a cohort earns after graduation, weighted by its declining share in the stock. A unit increase in V_t raises enrollment by $(1-\lambda)/d$. A permanent 1 percent increase in the relative wage raises V_t by $\beta^4/(1-\beta\lambda)$, so the long-run entry elasticity is

$$\eta^{LR} = \frac{(1-\lambda)\,\beta^4}{d\,(1-\beta\lambda)}. \tag{9}$$

Expectations. In the baseline, expectations are rational: E_t is the mathematical expectation conditional on the state in year t. Students then forecast how their peers' choices will affect future wages. Section 6 considers cobweb expectations instead, under which students expect the current wage to persist, $E_t w_{t+j} = w_t$. Then $V_t = \beta^4 w_t/(1-\beta\lambda)$, and

$$n_t = \eta\, w_t + \psi_t, \qquad \eta = \frac{(1-\lambda)\,\beta^4}{d\,(1-\beta\lambda)}. \tag{10}$$

Enrollment then moves only with the current wage, and students no longer anticipate that their peers' entry will depress future wages. Under the cobweb we estimate η in place of d.

2.2 Equilibrium and Solution

Definition 1. *A rational expectations competitive equilibrium is a collection of stochastic processes $\{w_t, N_t^d, N_t^h, n_t\}$ with bounded second moments such that (i) given $\{w_t\}$, $\{N_t^d\}$ solves the firm's problem (3); (ii) given $\{w_t\}$, $\{n_t, N_t^h\}$ solves the household's problem (6); and (iii) the market clears, $N_t^d = N_t^h \equiv N_t$.*

Enrollment depends on expected wages, which in turn depend on enrollment. The following result removes this fixed-point problem. Consider the planning problem

$$\max_{\{n_t\}} \; E_0 \sum_{t=0}^{\infty} \beta^t \Big[\theta_t N_t - \tfrac{f_2}{2} N_t^2 - \tfrac{d}{2} n_t^2 + d\,\psi_t\, n_t\Big] \quad \text{subject to} \quad (1)\text{–}(5), \tag{11}$$

whose objective is the sum of the firm's and the household's objectives once the wage bill and wage income cancel.

Proposition 1. *An allocation $\{N_t, n_t\}$ is a competitive equilibrium allocation if and only if it solves (11), with equilibrium wage $w_t = \theta_t - f_2 N_t$. The equilibrium is unique.*

The proof, in Appendix A.4, shows that the planner's first-order condition coincides with the household's once the firm's condition (4) is substituted for the wage.

Matrix form. Let the state and the control be

$$X_t = \big(N_t,\ n_{t-1},\ n_{t-2},\ n_{t-3},\ n_{t-4},\ \theta_t,\ \psi_t,\ \theta_{t-1},\ \psi_{t-1}\big)', \qquad u_t = n_t.$$

The constraints and the period return in (11) take the form

$$X_{t+1} = AX_t + Bu_t + C\varepsilon_{t+1}, \qquad r(X_t, u_t) = X_t' R X_t + Q u_t^2 + 2u_t W X_t, \tag{12}$$

with $\varepsilon_t = (\varepsilon_t^\theta, \varepsilon_t^\psi)'$ and the matrices given in Appendix A.5.

Solution. The value function is quadratic, $v(X) = X'PX + c$, and satisfies the Bellman equation

$$X'PX + c = \max_u \big\{ X'RX + Qu^2 + 2uWX + \beta\,(AX+Bu)'P(AX+Bu) + \beta\,\mathrm{tr}(C'PC) + \beta c \big\}. \tag{13}$$

The first-order condition gives the decision rule

$$u_t = -FX_t, \qquad F = (Q + \beta B'PB)^{-1}(W + \beta B'PA), \tag{14}$$

and matching the quadratic terms shows that P solves the algebraic Riccati equation

$$P = R + \beta A'PA - (W' + \beta A'PB)(Q + \beta B'PB)^{-1}(W + \beta B'PA). \tag{15}$$

The second-order condition holds: the block of P for the endogenous states is negative semidefinite because the objective is concave in them, and B selects only n_{t-1}, so $Q + \beta B'PB \le -d/2 < 0$. Because F does not depend on C, certainty equivalence holds. We solve (15) with a standard solver after rescaling (A, B) by $\sqrt{\beta}$. Substituting (14) into (12) gives the equilibrium law of motion

$$X_{t+1} = MX_t + C\varepsilon_{t+1}, \qquad M = A - BF. \tag{16}$$

Career value. Let G_w be the row vector with $w_t = G_w X_t$. Since $E_t X_{t+k} = M^k X_t$, the career value in (8) is

$$V_t = \beta^4 \sum_{j=0}^{\infty} (\beta\lambda)^j\, G_w M^{4+j} X_t = \beta^4\, G_w M^4 \big(I - \beta\lambda M\big)^{-1} X_t. \tag{17}$$

Although V_t forecasts wages four or more years ahead, it is linear in the current state, so any shock moves V_t and enrollment at once. Under cobweb expectations no planning problem is needed: by (4) and (10), $n_t = -\eta f_2 N_t + \eta\,\theta_t + \psi_t \equiv KX_t$, and $X_{t+1} = (A + BK)X_t + C\varepsilon_{t+1}$. Under either form of expectations, w_t, N_t, and π_t are linear functions of X_t.

3 Data

We use annual U.S. data for 1968–2010, 43 observations in all. The relative wage and the relative stock come from CPS ASEC microdata harmonized by IPUMS (Flood et al. 2025). Engineers are employed workers in OCC1990 codes 044–059, and college graduates are persons with four or more years of college (EDUC $\geq$ 110). For each year we compute the median annual earnings of the two groups, W_t^E and W_t^C, and their weighted counts, L_t^E and L_t^C.[3] The degree share uses the numbers of engineering and of all bachelor's degrees, D_t^E and D_t^A, from National Science Foundation, National Center for Science and Engineering Statistics (2013), Table 5. The NSF did not publish degrees by field for 1999, and we interpolate that year linearly. The observed series are

$$\hat{w}_t = \log \frac{W_t^E}{W_t^C}, \qquad \hat{N}_t = \log \frac{L_t^E}{L_t^C}, \qquad \hat{\pi}_t = \log \frac{D_t^E}{D_t^A}. \tag{18}$$

Taking ratios removes factors common to both groups, such as inflation and the growth of the college-educated population. The three series have different trends: the relative stock falls by about 2.6 log points a year and the degree share by about 0.5, while the relative wage has almost none. We therefore remove a separate linear trend from each series,

$$x_t = a_x + b_x\, t + \tilde{x}_t, \qquad x \in \{\hat{w}, \hat{N}, \hat{\pi}\}, \tag{19}$$

and use the residuals $\tilde{w}_t$, $\tilde{N}_t$, and $\tilde{\pi}_t$ as observations. Figure 1 plots the series. After detrending, the degree share fluctuates the most: it lies 33 log points below trend in 1976 and 35 log points above trend in 1985.

4 Estimation

4.1 The Likelihood Function

Observation equation. Let $y_t = (\tilde{w}_t, \tilde{N}_t, \tilde{\pi}_t)'$. Each observation equals its model counterpart, GX_t, plus a measurement error:

$$y_t = GX_t + v_t, \qquad v_{t+1}^x = D_x\, v_t^x + \sigma_x\, e_{t+1}^x, \quad x \in \{w, N, \pi\}, \tag{20}$$

[3] Years are survey years; the CPS ASEC asks about earnings in the previous calendar year.

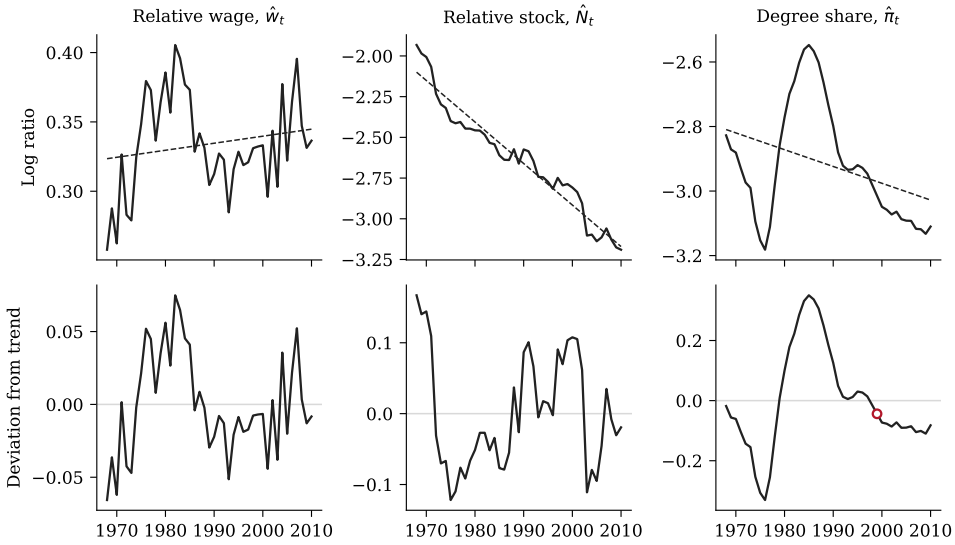


Figure 1: Relative wage, relative stock, and degree share, 1968–2010

Notes: Top: the log ratios in (18) (solid) and their linear trends (dashed). Bottom: the detrended series. The open circle marks the interpolated value for 1999. *Sources:* CPS ASEC (IPUMS); NSF 13-327, Table 5.

where the e_t^x are standard normal and independent of each other and of the structural shocks. Measurement error in the CPS series is not negligible. The survey contains only 611 to 1,409 engineers a year, and the measured stock falls by 17.7 log points in 2003, when the CPS adopted a new occupational classification (U.S. Bureau of Labor Statistics n.d.), although new graduates add only about 4 percent to the stock each year. Because about half of the CPS sample overlaps between adjacent years, we allow these errors to be serially correlated. The degree share comes from administrative records, and we set $D_\pi = 0$.

Likelihood. Adding the measurement errors to the state, $s_t = (X_t', v_t')'$, the law of motion (16) and the observation equation (20) form the state-space system

$$s_{t+1} = \mathcal{M} s_t + \mathcal{C}\epsilon_{t+1}, \qquad y_t = \mathcal{G} s_t, \tag{21}$$

where ϵ_t collects the two structural shocks and the three measurement innovations (Appendix B.1). Let $\hat{s}_t = E[s_t \mid y^{t-1}]$ with $y^{t-1} = (y_1, \ldots, y_{t-1})$. The Kalman filter gives the innovations representation

$$\hat{s}_{t+1} = \mathcal{M}\hat{s}_t + K_t a_t, \qquad y_t = \mathcal{G}\hat{s}_t + a_t, \qquad a_t \mid y^{t-1} \sim N(0, \Omega_t), \tag{22}$$

where the innovation a_t is the part of y_t that cannot be predicted from past observations, and the gain K_t and covariance Ω_t follow from the filtering recursions started at the stationary distribution (Appendix B.2). The innovations are serially independent and in one-to-one correspondence with the observations, so the joint density of the sample is the product of the innovation densities (Appendix B.3):

$$\log L = \sum_{t=1}^{T} \Big[-\tfrac{3}{2}\log 2\pi - \tfrac{1}{2}\log\det\Omega_t - \tfrac{1}{2}\, a_t'\Omega_t^{-1} a_t \Big]. \tag{23}$$

Estimation. The parameter vector is

$$\Theta = \big(f_2,\, d,\, \lambda,\, a_1^{\theta},\, a_2^{\theta},\, \sigma_\theta,\, a_1^{\psi},\, a_2^{\psi},\, \sigma_\psi,\, \sigma_w,\, D_w,\, \sigma_N,\, D_N,\, \sigma_\pi\big)' \in \mathcal{P} \subset \mathbb{R}^{14}, \tag{24}$$

where $\mathcal{P}$ requires f_2, d, and the standard deviations to be positive, $\lambda \in (0,1)$, $|D_w|, |D_N| < 1$, and both autoregressions to be stationary. The estimator is $\hat{\Theta} = \arg\max_{\Theta\in\mathcal{P}} \log L(\Theta)$. Evaluating $\log L(\Theta)$ takes three steps, listed in Algorithm 1: solve the planning problem for M, build the state-space system (21), and run the Kalman filter. If the Riccati equation has no solution, $\mathcal{M}$ is not stable, or some Ω_t is not positive definite, we set $\log L = -10^{10}$.

Algorithm 1 Evaluating the log likelihood $\log L(\Theta)$

Input: parameters Θ; observations $y_1, \ldots, y_T$

1: Construct A, B, C, R, Q, W from Θ; solve (15) for P; set F by (14) and $M = A - BF$

2: Construct $\mathcal{M}, \mathcal{C}, \mathcal{G}$; set $\hat{s}_1 = 0$, $\Sigma_1 = \mathcal{M}\Sigma_1\mathcal{M}' + \mathcal{C}\mathcal{C}'$, and $\log L \leftarrow 0$

3: **for** $t = 1, \ldots, T$ **do**

4: $\quad a_t \leftarrow y_t - \mathcal{G}\hat{s}_t, \quad \Omega_t \leftarrow \mathcal{G}\Sigma_t\mathcal{G}', \quad \log L \leftarrow \log L - \tfrac{3}{2}\log 2\pi - \tfrac{1}{2}\log\det\Omega_t - \tfrac{1}{2}\, a_t'\Omega_t^{-1}a_t$

5: $\quad K_t \leftarrow \mathcal{M}\Sigma_t\mathcal{G}'\Omega_t^{-1}, \quad \hat{s}_{t+1} \leftarrow \mathcal{M}\hat{s}_t + K_t a_t, \quad \Sigma_{t+1} \leftarrow (\mathcal{M} - K_t\mathcal{G})\Sigma_t(\mathcal{M} - K_t\mathcal{G})' + \mathcal{C}\mathcal{C}'$

6: **end for**

7: **return** $\log L$

Optimization and inference. We impose $\mathcal{P}$ by reparametrization: f_2, d, and the standard deviations are exponentials of unconstrained parameters, λ is a logistic function, a_2, D_w, and D_N are hyperbolic tangents, and $a_1 = (1 - a_2)\tanh(z)$, which keeps each autoregression stationary. We maximize with the Nelder–Mead method, then BFGS, then

Nelder–Mead again from the BFGS solution. From eight starting values, seven reach the same estimate; the eighth stops at a lower local maximum ($\log L = 259.45$). Standard errors come from the inverse of the numerical Hessian; σ_π, estimated on the boundary of $\mathcal{P}$, is held fixed. Because the likelihood can be asymmetric, we also report profile likelihood intervals: fixing one parameter at x and re-maximizing over the other 13 gives $\log L_p(x)$, and the values of x with

$$LR(x) = 2\left[\log L(\hat{\Theta}) - \log L_p(x)\right] \leq 3.84 \tag{25}$$

form a 95 percent confidence interval.

4.2 Estimates

Table 1 reports the estimates, and Figure 2 plots $LR(x)$ for nine parameters.

Demand. The estimate of f_2 is 0.64, so the demand elasticity is $1/f_2 = 1.57$, within the range of 1.2 to 2.2 found by Ryoo & Rosen (2004). The profile likelihood bounds f_2 only from below, at 0.135: the data rule out a nearly flat demand curve, with an elasticity above about 7.4, but not a steep one. An interval of two standard errors would be misleading here, since it includes negative values of f_2 that the profile likelihood rejects.

Supply. The coefficient on V_t, $(1-\lambda)/d$, is 0.59: a one-percentage-point increase in career value raises the enrollment share by 0.59 percent. A permanent 1 percent wage increase raises V_t by $\beta^4/(1-\beta\lambda) = 10.26$ points, so the long-run entry elasticity is $0.59 \times 10.26 = 6.07$, above the range of 2.5 to 4.5 that Ryoo & Rosen (2004) report for the long-run supply elasticity with respect to career prospects, though imprecisely estimated. The enrollment cost d is bounded only from above, at 0.44: the data rule out enrollment that is insensitive to wages, but not enrollment that is highly sensitive.

Stock. The estimate of λ is 0.955. Since $1 - \lambda$ is the share of new graduates in the stock, the average ratio of engineering degrees to engineers over 1968–2010, 0.041, gives an accounting benchmark of $\lambda = 0.959$. The estimate, which uses only the dynamics of the three series, is almost identical.

Table 1: Parameter estimates

Parameter	Description	Estimate	S.E.	Profile 95% interval	
				Lower	Upper
A. Demand, enrollment, and stock					
f_2	Slope of inverse demand	0.638	(0.487)	0.135	> 6.32
d	Enrollment cost slope	0.075	(0.054)	< 0.009	0.439
λ	Share of incumbents in the stock	0.955	(0.029)	0.895	> 0.995
B. Demand shock θ					
a_1^θ	AR(2) coefficient	0.697	(0.319)	-0.090	> 1.656
a_2^θ	AR(2) coefficient	0.176	(0.324)	< -0.797	0.646
σ_θ	Innovation s.d.	0.0137	(0.0049)	< 0.005	0.030
C. Preference shock ψ					
a_1^ψ	AR(2) coefficient	1.735	(0.086)	1.541	1.876
a_2^ψ	AR(2) coefficient	-0.815	(0.086)	-0.954	-0.618
σ_ψ	Innovation s.d.	0.0329	(0.0072)	0.022	0.053
D. Measurement error					
σ_w	Wage: innovation s.d.	0.0180	(0.0038)	< 0.010	0.026
D_w	Wage: persistence	-0.098	(0.267)	< -0.900	> 0.704
σ_N	Stock: innovation s.d.	0.0492	(0.0053)	0.040	0.062
D_N	Stock: persistence	0.729	(0.108)	0.507	0.931
σ_π	Degree share: s.d.	0	—	0	0.009
E. Implied quantities					
$1/f_2$	Demand elasticity	1.57	(1.20)		
$(1-\lambda)/d$	Coefficient on V_t	0.59	(0.38)		
η^{LR}	Long-run entry elasticity	6.07	(3.63)		
	Preference shock: modulus, period	0.90, 22.3 yr			
$\log L$	Maximized log likelihood	260.47			

Notes: $T = 43$; $\beta = 0.96$ is fixed. Standard errors are from the inverse numerical Hessian, with σ_π held at its boundary value; those of implied quantities use the delta method. Profile intervals contain the values with $LR(x) \leq 3.84$; ">" and "<" mean that the bound is not reached within the grid.

Shocks. The demand shock has real roots, 0.89 and -0.20, and its parameters are poorly determined. The preference shock is precisely estimated, with profile intervals closed on both sides. Its roots are a complex pair, $0.867 \pm 0.251i$, with modulus 0.90: a damped

oscillation with a period of about 22 years. Restricting both shocks to real roots lowers the log likelihood to 258.11. The likelihood ratio statistic, 4.72, exceeds the 5 percent critical value of 3.84, so the restriction is rejected.[4]

Measurement error. The stock error has a standard deviation of 0.049 and persistence 0.73, against a standard deviation of 0.079 for the detrended stock itself: the stock series is noisy. The wage error has a standard deviation of 0.018, and the data cannot tell whether it is serially correlated. The degree-share error is estimated at zero, on the boundary of the parameter space.[5]

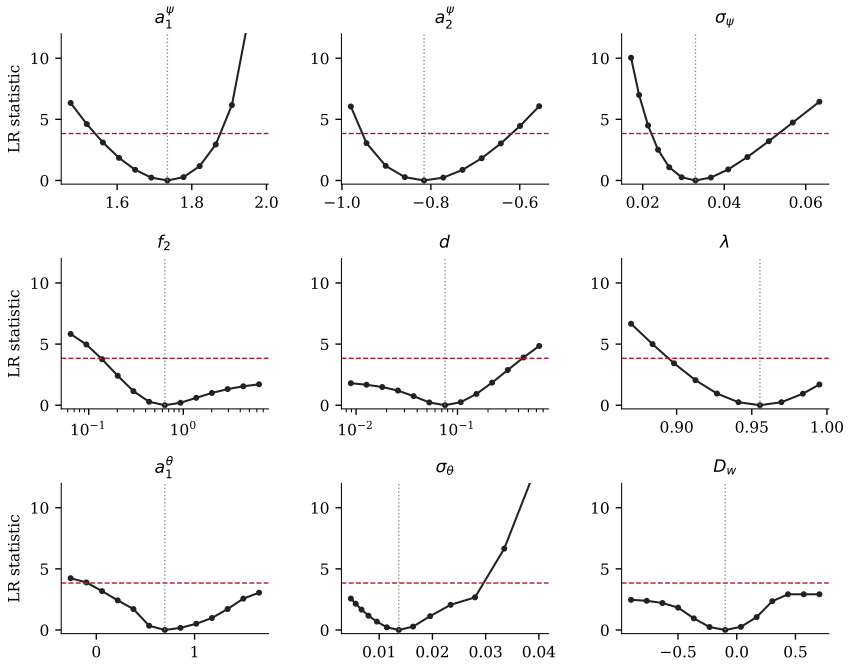


Figure 2: Profile likelihood for selected parameters

Notes: Each panel plots $LR(x)$ in (25). Rows: parameters bounded on both sides; bounded on one side; poorly determined. The dashed line is the 5 percent critical value, 3.84; the dotted line marks the estimate. The axes for f_2 and d are logarithmic.

[4]Because the null hypothesis is an inequality, $(a_1^\psi)^2 + 4a_2^\psi \geq 0$, its least favorable point lies on the boundary of the real-root region, where the statistic follows an equal mixture of χ_0^2 and χ_1^2 with a 5 percent critical value of 2.71. The conclusion is unchanged.

[5]A zero estimate of a measurement-error variance is known in factor analysis as a Heywood case.

4.3 Model Fit

The parameters maximize the likelihood rather than match particular moments. To check the fit, we simulate 1,000 samples of 43 years from the estimated model, including measurement error, detrend each as in (19), and compare the statistics in the data with their distribution across samples (Table 2).

Table 2: Data and model moments

		Model			
	Data	Population	Median	5%	95%
Standard deviation					
$\tilde{w}_t$	0.035	0.037	0.028	0.021	0.040
$\tilde{N}_t$	0.079	0.080	0.064	0.046	0.088
$\tilde{\pi}_t$	0.169	0.140	0.118	0.065	0.195
First-order autocorrelation					
$\tilde{w}_t$	0.569	0.653	0.431	0.016	0.725
$\tilde{N}_t$	0.765	0.778	0.657	0.408	0.816
$\tilde{\pi}_t$	0.957	0.947	0.934	0.869	0.962
Correlation of the wage with the future degree share					
$\text{corr}(\tilde{w}_t, \tilde{\pi}_{t+4})$	0.750	0.286	0.352	−0.118	0.691
$\text{corr}(\tilde{w}_t, \tilde{\pi}_{t+5})$	0.763	0.319	0.365	−0.072	0.711

Notes: Population moments are from the stationary distribution of the model (Appendix C). The median and percentiles are across 1,000 simulated 43-year samples, each including measurement error and detrended as in (19).

The model reproduces the volatility and persistence of all three series: each standard deviation and first-order autocorrelation in the data lies within the 5th–95th percentile range of the simulated samples. What the model has difficulty matching is the correlation between the wage and the future degree share. In the data, $\text{corr}(\tilde{w}_t, \tilde{\pi}_{t+4}) = 0.75$: years with high wages see more students enroll and, four years later, more graduates. The model reproduces the shape of the correlation function (Figure 4), but for leads of three to six years the data lie above the 95th percentile of the simulated samples. Section 5.1 shows that this gap follows from rational expectations.

5 Dynamics

5.1 Impulse Responses

In year 0 the economy is hit by a one-standard-deviation demand or preference shock, and no further shocks occur. By (16), the state h years after shock j is $X_h = M^h C e_j$, where e_j is the jth unit vector, and the wage and enrollment are $w_h = G_w X_h$ and $n_h = -F X_h$. Figure 3 plots the responses.

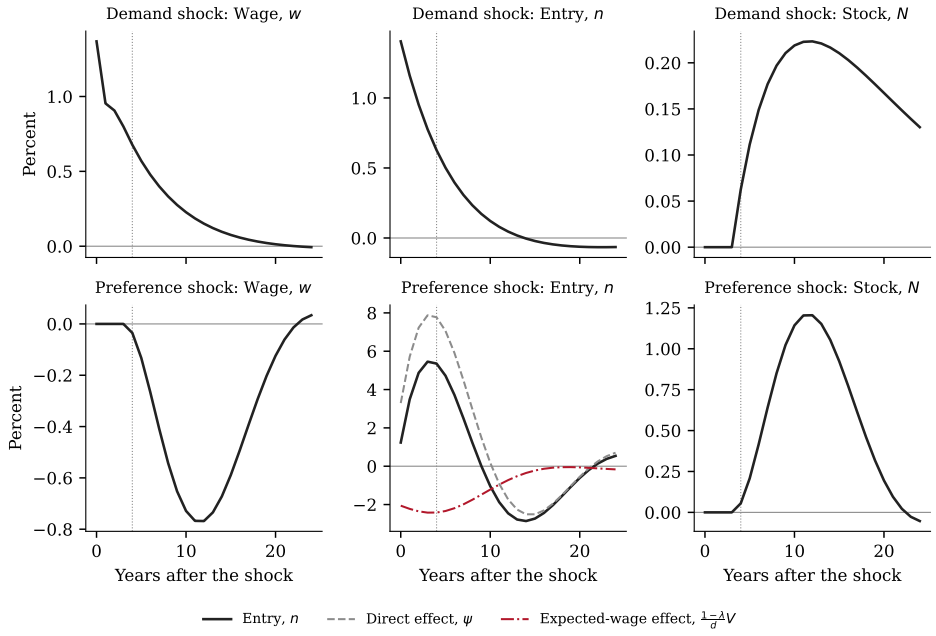


Figure 3: Responses to one-standard-deviation demand and preference shocks

Notes: Top row: demand shock; bottom row: preference shock. Columns: percentage deviations of the relative wage, the enrollment share, and the relative stock. The middle bottom panel splits enrollment into the direct effect of preferences, ψ_h (dashed), and the effect of expected wages, $\frac{1-\lambda}{d} V_h$ (dash-dotted). The vertical dotted line marks year 4, when the year-0 cohort graduates.

There are two main points to take away from Figure 3. First, under rational expectations students look ahead and do not chase temporarily high wages. A demand shock raises the relative wage by 1.37 percent on impact, and enrollment rises by 1.40 percent in the same year, long before the higher wage reaches graduates. The wage increase, however, decays by half in about four years. Students anticipate this, so the career value V_t rises by only 2.4

percentage points, compared with 14.0 points if the same wage increase were permanent, and enrollment responds only about one-sixth as much. This is why the model understates the correlation between the wage and the future degree share: wage fluctuations induce only small changes in enrollment, so the degree share moves mainly with preferences that are unrelated to wages.

Second, the preference shock is largely self-offsetting. The shock is 3.29 percent on impact, but enrollment rises by only 1.24 percent, about 37 percent of the shock. Students foresee that they and the cohorts that follow will enter the field together, so that four years later the stock will rise and the wage will fall. The career value therefore falls at once and offsets about 63 percent of the direct effect. The preference shock itself oscillates with a period of about 22 years; the stock peaks at 1.20 percent in years 11–12, when the wage reaches its trough of −0.77 percent. Under both shocks the stock adjusts slowly: without new entry, a deviation of the stock from steady state has a half-life of 15.2 years, so the effects of a single shock last for more than a decade.

5.2 Variance Decomposition

The five shocks are independent, so the variance of each series can be split by shock (Appendix C). Table 3 reports the result.

Table 3: Variance decomposition

	Observed series			Model variables		
	$\tilde{w}_t$	$\tilde{N}_t$	$\tilde{\pi}_t$	w_t	N_t	n_t
Demand shock	0.42	0.01	0.03	0.56	0.07	0.03
Preference shock	0.34	0.18	0.97	0.44	0.93	0.97
Measurement error	0.24	0.81	0.00			

Notes: Shares of the stationary variance contributed by each shock; each column sums to one. Measurement error is the series' own. The model variables exclude measurement error.

Three results stand out. First, about 97 percent of the variance of enrollment and of the degree share comes from the preference shock. Consistent with Section 5.1, the large swings in the choice of major come mainly from changes in the attractiveness of

engineering, not from wages. Second, 81 percent of the variance of the observed stock comes from measurement error. Third, both shocks matter for wages: the demand shock accounts for 56 percent of the variance of the true wage and the preference shock for 44 percent.

6 Rational Expectations versus the Cobweb

This section reports estimates under cobweb expectations (10), with the same data and likelihood. The cobweb model replaces d with η and also has 14 free parameters.[6]

Table 4 reports the two sets of estimates. The models fit the data about equally well: the cobweb log likelihood is 1.5 higher, and since the models are not nested but have the same number of parameters, a gap of this size does not discriminate between them. The estimates of λ and of the stock measurement error are almost identical, and so is the conclusion that enrollment is highly sensitive to wages: $\eta = 5.10$ under the cobweb against $\eta^{LR} = 6.07$ under rational expectations. The differences lie on the demand side. The cobweb model implies a demand elasticity of about 6.0, a nearly flat demand curve, against 1.57. Its demand shock is a slowly decaying, almost deterministic long cycle: the roots have modulus 0.97 and a period of about 25 years, and the innovation standard deviation is one-seventh of the rational-expectations estimate.

Figure 4 shows that the cobweb model matches the correlation between the wage and the future degree share. At $k = 4$, its simulated median is 0.56 and its 5th–95th percentile range, $[0.14, 0.83]$, covers the value of 0.75 in the data. Students under the cobweb do not distinguish temporary from permanent wage changes: a 1 percent increase in the current wage raises enrollment by 5.1 percent on impact, against about 1.0 percent under rational expectations when the same increase comes from a demand shock. Enrollment follows the current wage closely, so the degree share four years later is strongly correlated with today's

[6]The cobweb likelihood has several local maxima. Of six starting values, two reach a solution with $\log L = 259.59$ and $\eta = 0.41$; two reach $\log L = 263.07$, at which $\sigma_\psi \to 0$ and the roots of the preference shock reach the unit circle, so that the shock becomes a deterministic sine wave and the model has no stationary distribution; and two reach $\log L = 261.92$ and 262.00 with $\eta \approx 5.1$. We exclude the degenerate solution and report the highest maximum among solutions whose roots all have modulus below 0.99.

Table 4: Rational-expectations and cobweb estimates

	Rational expectations		Cobweb	
	Estimate	S.E.	Estimate	S.E.
$\log L$	260.47		262.00	
f_2	0.638	(0.487)	0.167	(0.246)
d	0.075	(0.054)		
η			5.10	(1.32)
λ	0.955	(0.029)	0.946	(0.031)
a_1^θ	0.697	(0.319)	1.877	(0.084)
a_2^θ	0.176	(0.324)	−0.936	(0.081)
σ_θ	0.0137	(0.0049)	0.0019	(0.0010)
a_1^ψ	1.735	(0.086)	1.587	(0.142)
a_2^ψ	−0.815	(0.086)	−0.732	(0.144)
σ_ψ	0.0329	(0.0072)	0.0217	(0.0046)
σ_w, D_w	0.0180, −0.098		0.0214, −0.052	
σ_N, D_N	0.0492, 0.729		0.0491, 0.729	
σ_π	0		0.0037	
Demand elasticity $1/f_2$	1.57		5.99	
Entry elasticity	6.07		5.10	
Demand shock: modulus, period	0.89, —		0.97, 25.5 yr	
Preference shock: modulus, period	0.90, 22.3 yr		0.86, 16.4 yr	

Notes: The entry elasticity is η^{LR} under rational expectations and η under the cobweb. Under rational expectations the demand shock has real roots; the larger is reported, and there is no period. Standard errors of the measurement-error parameters are omitted.

wage.

Figure 5 compares the impulse responses. Under a demand shock, the cobweb model produces the classic cobweb cycle: the wage rises, enrollment increases sharply, too many students graduate four years later, the wage falls back, and enrollment drops below normal. Part of this reversal, however, comes from the cycle in the demand shock itself. Under rational expectations, enrollment responds at once and then declines smoothly. Under a preference shock, cobweb enrollment moves one for one with the shock for four years, because the wage has not yet changed, whereas under rational expectations enrollment

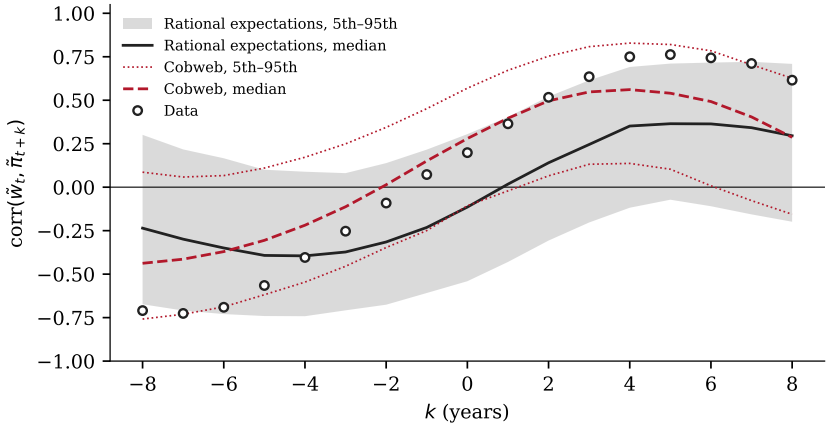


Figure 4: Correlation of the wage with the degree share k years later

Notes: Circles: data. Solid line and shaded band: median and 5th–95th percentile range across 1,000 simulated 43-year samples from the rational-expectations model. Dashed and dotted lines: the same for the cobweb model. Each sample is detrended as in the data.

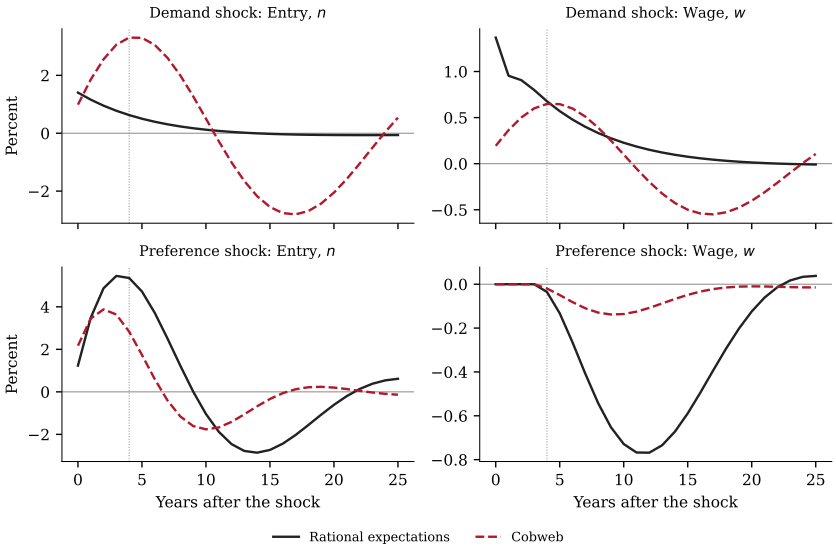


Figure 5: Responses under rational expectations and the cobweb

Notes: Solid: rational expectations; dashed: cobweb. Each shock is one standard deviation as estimated in that model. The vertical dotted line marks year 4.

rises on impact by only about 37 percent of the shock.

7 Conclusion

Students choose a major four years before they enter the labor market, and what they care about is the wage over their careers. Estimating the Ryoo–Rosen model jointly by Kalman-filter maximum likelihood, we find that enrollment is highly sensitive to persistent wage changes under both rational and cobweb expectations. The large swings in the degree share are nonetheless not driven mainly by wages: about 97 percent of their variance comes from changes in the attractiveness of engineering itself.

What the data cannot settle is whether students look through temporary wage changes. Since Freeman (1971, 1976), booms and busts in the market for engineers have often been explained as a cobweb cycle in which they do not, and our data do not reject this explanation; Ryoo & Rosen (2004, p. S135) similarly found it difficult "to isolate the precise expectations mechanism." Under rational expectations, students discount a wage increase they expect to fade, and the model understates the link between wages and future degrees. The cobweb model matches this link and attains a similar likelihood, but only with a nearly flat demand curve and an almost deterministic long cycle in demand. The two forms of expectations agree that enrollment responds strongly to persistent wage changes, but their estimates of the demand elasticity differ by a factor of almost four.

Our numbers rest on several fixed choices: the discount factor is set at 0.96, each series is linearly detrended before estimation, the two shocks are independent, and enrollment involves no adjustment costs. The sample has only 43 annual observations, which bound the demand elasticity and the slope of the enrollment cost only from above, and about 81 percent of the variance of the observed stock is measurement error. These choices define the scope of our conclusions.

The framework itself is not specific to engineers. Occupations in which entry requires years of training and the return to entry depends on wages earned long afterward, such as physicians, lawyers, teachers, and scientists, share its structure. Given series on relative wages, stocks, and entry, the planning problem, the state-space representation, and the likelihood carry over, with the state vector adjusted to the length of training. Estimating the model across occupations would show whether the patterns found here are particular to engineering or common to markets for trained labor.

References

Altug, S. (1989), 'Time-to-build and aggregate fluctuations: Some new evidence', *International Economic Review* **30**(4), 889–920.

Arrow, K. J. & Capron, W. M. (1959), 'Dynamic shortages and price rises: The engineer-scientist case', *Quarterly Journal of Economics* **73**(2), 292–308.

Flood, S., King, M., Rodgers, R., Ruggles, S., Warren, J. R. et al. (2025), 'IPUMS CPS: Version 13.0 [dataset]', Minneapolis, MN: IPUMS. https://doi.org/10.18128/D030.V13.0.

Freeman, R. B. (1971), *The Market for College-Trained Manpower: A Study in the Economics of Career Choice*, Harvard University Press, Cambridge, MA.

Freeman, R. B. (1975), 'Supply and salary adjustments to the changing science manpower market: Physics, 1948–1973', *American Economic Review* **65**(1), 27–39.

Freeman, R. B. (1976), 'A cobweb model of the supply and starting salary of new engineers', *Industrial and Labor Relations Review* **29**(2), 236–248.

Hall, G. J. (1996), 'Overtime, effort, and the propagation of business cycle shocks', *Journal of Monetary Economics* **38**(1), 139–160.

Hansen, L. P. & Sargent, T. J. (1980), 'Formulating and estimating dynamic linear rational expectations models', *Journal of Economic Dynamics and Control* **2**, 7–46.

Hansen, L. P. & Sargent, T. J. (2013), *Recursive Models of Dynamic Linear Economies*, Princeton University Press, Princeton, NJ.

Ireland, P. N. (2004), 'A method for taking models to the data', *Journal of Economic Dynamics and Control* **28**(6), 1205–1226.

National Science Foundation, National Center for Science and Engineering Statistics (2013), 'Science and engineering degrees: 1966–2010', Detailed Statistical Tables NSF 13-327, Arlington, VA.

Ryoo, J. & Rosen, S. (2004), 'The engineering labor market', *Journal of Political Economy* **112**(S1), S110–S140.

Sargent, T. J. (1978), 'Estimation of dynamic labor demand schedules under rational expectations', *Journal of Political Economy* **86**(6), 1009–1044.

Sargent, T. J. (1989), 'Two models of measurements and the investment accelerator', *Journal of Political Economy* **97**(2), 251–287.

Siow, A. (1984), 'Occupational choice under uncertainty', *Econometrica* **52**(3), 631–646.

U.S. Bureau of Labor Statistics (n.d.), 'Occupation and industry classification systems used in the Current Population Survey', https://www.bls.gov/cps/cpsoccind.htm. Accessed October 21, 2026.

Zarkin, G. A. (1985), 'Occupational choice: An application to the market for public school teachers', *Quarterly Journal of Economics* **100**(2), 409–446.

A Derivations for the Model

A.1 Stock Accumulation

Divide $L_{t+1} = \delta L_t + S_{t-3}$ by L_t. On a balanced growth path, $L_{t+1}/L_t = 1+g$ and $S_{t-3}/L_t = \bar{s}$, so $1+g = \delta + \bar{s}$. Let $\hat{\ell}_t$ and $\hat{s}_t$ be the log deviations of L_t and S_t from the path. A first-order expansion of $(1+g)e^{\hat{\ell}_{t+1}-\hat{\ell}_t} = \delta + \bar{s}\, e^{\hat{s}_{t-3}-\hat{\ell}_t}$ gives

$$(1+g)(\hat{\ell}_{t+1}-\hat{\ell}_t) = \bar{s}\,(\hat{s}_{t-3}-\hat{\ell}_t) \implies \hat{\ell}_{t+1} = \lambda\,\hat{\ell}_t + (1-\lambda)\,\hat{s}_{t-3}, \qquad \lambda = \frac{\delta}{1+g}.$$

With the same equation for all college graduates (superscript c), $N_t = \hat{\ell}_t - \hat{\ell}^c_t$, and $n_t = \hat{s}_t - \hat{s}^c_t$,

$$N_{t+1} = \lambda N_t + (1-\lambda)\,n_{t-3} + (\lambda - \lambda^c)\big(\hat{\ell}^c_t - \hat{s}^c_{t-3}\big).$$

In our data the detrended standard deviation of the last term is 0.0022, against 0.0787 for N_t; dropping it gives (5).

A.2 The Household's Problem

Iterating the constraint in (6), $\partial N_s^h/\partial n_t = (1-\lambda)\lambda^{s-t-4}$ for $s \geq t+4$ and zero otherwise. Since the household takes wages as given, the derivative of its objective with respect to n_t, divided by β^t, is

$$-d\,(n_t - \psi_t) + (1-\lambda)\,E_t \sum_{j=0}^{\infty} \beta^{4+j}\lambda^j\, w_{t+4+j} = 0,$$

which is (7); the second derivative is $-d < 0$. If w_{t+k} rises by one unit for all $k \geq 0$, then $\Delta V_t = \beta^4 \sum_{j\geq 0}(\beta\lambda)^j = \beta^4/(1-\beta\lambda)$ and $\Delta n_t = \eta^{LR}$.

A.3 Second-Order Autoregressions with Complex Roots

The response $x_h = \partial x_{t+h}/\partial \varepsilon_t$ of $x_t = a_1 x_{t-1} + a_2 x_{t-2} + \varepsilon_t$ satisfies $x_h = a_1 x_{h-1} + a_2 x_{h-2}$ with $x_0 = 1$, $x_{-1} = 0$. With distinct roots $z_{1,2}$ of $z^2 = a_1 z + a_2$, the solution is $x_h = (z_1^{h+1} - z_2^{h+1})/(z_1 - z_2)$. If $a_1^2 + 4a_2 < 0$, then $z_{1,2} = re^{\pm i\omega}$ with $r^2 = z_1 z_2 = -a_2$ and $r\cos\omega = a_1/2$, and

$$x_h = \frac{2i\, r^{h+1}\sin\big((h+1)\omega\big)}{2i\, r\sin\omega} = r^h\,\frac{\sin\big((h+1)\omega\big)}{\sin\omega},$$

an oscillation with period $2\pi/\omega$ whose amplitude decays at rate r.

A.4 Proof of Proposition 1

Since $\partial(\theta_s N_s - \frac{f_2}{2}N_s^2)/\partial N_s = \theta_s - f_2 N_s$, the planner's first-order condition for n_t is

$$d\,(n_t - \psi_t) = (1-\lambda)\,E_t \sum_{j=0}^{\infty} \beta^{4+j}\lambda^j\,\big(\theta_{t+4+j} - f_2 N_{t+4+j}\big). \tag{26}$$

For $f_2, d > 0$ the objective is strictly concave and the constraints are linear, and for processes with bounded second moments $\beta < 1$ ensures the transversality condition. Hence $\{N_t, n_t\}$ solves (11) if and only if (26) holds for all t, and the solution is unique.

Necessity. In an equilibrium, substituting the firm's condition $w_s = \theta_s - f_2 N_s$ into the household's condition (7) gives (26).

Sufficiency. Let $\{N_t, n_t\}$ satisfy (26) and set $w_s = \theta_s - f_2 N_s$, $N_s^d = N_s^h = N_s$. Then (4) and market clearing hold by construction, and (26) becomes (7). Both private problems

are strictly concave, so these first-order conditions are sufficient, and the allocation with this wage is an equilibrium. Uniqueness follows from that of the planner's solution. □

A.5 Linear-Quadratic Representation

With X_t and u_t as in Section 2.2, the matrices in (12) are

$$A = \begin{pmatrix} \lambda & 0 & 0 & 1-\lambda & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & a_1^\theta & 0 & a_2^\theta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & a_1^\psi & 0 & a_2^\psi \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \sigma_\theta & 0 \\ 0 & \sigma_\psi \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The only nonzero elements of R are $R_{11} = -f_2/2$ and $R_{16} = R_{61} = 1/2$; $Q = -d/2$; and W has $d/2$ in the position of ψ_t and zeros elsewhere. These reproduce the period return in (11).

B The Kalman Filter and the Likelihood

B.1 The State-Space System

With $s_t = (X_t', v_t')'$ and $\epsilon_t = (\varepsilon_t^\theta, \varepsilon_t^\psi, e_t^w, e_t^N, e_t^\pi)' \sim$ i.i.d. $N(0, I_5)$, the matrices in (21) are

$$\mathcal{M} = \begin{pmatrix} M & 0 \\ 0 & D \end{pmatrix}, \qquad \mathcal{C} = \begin{pmatrix} C & 0 \\ 0 & \Sigma_v \end{pmatrix}, \qquad \mathcal{G} = \begin{pmatrix} G & I_3 \end{pmatrix},$$

where $D = \mathrm{diag}(D_w, D_N, 0)$, $\Sigma_v = \mathrm{diag}(\sigma_w, \sigma_N, \sigma_\pi)$, and the rows of G select $w_t = -f_2 N_t + \theta_t$, N_t, and $\pi_t = n_{t-4}$.

B.2 The Filtering Recursions

Let $\Sigma_t = \mathrm{Var}(s_t \mid y^{t-1})$. Suppose $s_t \mid y^{t-1} \sim N(\hat{s}_t, \Sigma_t)$, which holds at $t = 1$ with $\hat{s}_1 = 0$ and Σ_1 the stationary variance, $\Sigma_1 = \mathcal{M}\Sigma_1\mathcal{M}' + \mathcal{C}\mathcal{C}'$. Then $y_t \mid y^{t-1} \sim N(\mathcal{G}\hat{s}_t, \Omega_t)$ with

$\Omega_t = \mathcal{G}\Sigma_t\mathcal{G}'$, and, conditional on y^{t-1},

$$E[s_{t+1}] = \mathcal{M}\hat{s}_t, \qquad \mathrm{Var}(s_{t+1}) = \mathcal{M}\Sigma_t\mathcal{M}' + \mathcal{C}\mathcal{C}', \qquad \mathrm{Cov}(s_{t+1}, y_t) = \mathcal{M}\Sigma_t\mathcal{G}'.$$

Conditioning further on y_t with the formulas for jointly normal vectors gives

$$\hat{s}_{t+1} = \mathcal{M}\hat{s}_t + K_t a_t, \qquad K_t = \mathcal{M}\Sigma_t\mathcal{G}'\,\Omega_t^{-1}, \qquad a_t = y_t - \mathcal{G}\hat{s}_t,$$
$$\Sigma_{t+1} = \mathcal{M}\Sigma_t\mathcal{M}' + \mathcal{C}\mathcal{C}' - K_t\Omega_t K_t' = (\mathcal{M} - K_t\mathcal{G})\,\Sigma_t\,(\mathcal{M} - K_t\mathcal{G})' + \mathcal{C}\mathcal{C}',$$

which completes the induction and yields (22). The last form, a sum of positive semidefinite matrices, is the one used in Algorithm 1.

B.3 The Prediction Error Decomposition

For $r < t$, a_r is a function of y^{t-1} and $E[a_t \mid y^{t-1}] = 0$, so $E[a_t a_r'] = 0$; being normal, the innovations are independent. The map from y^T to a^T is triangular with unit Jacobian. Hence

$$f(y_1, \ldots, y_T) = \prod_{t=1}^{T} f\big(y_t \mid y^{t-1}\big) = \prod_{t=1}^{T} (2\pi)^{-3/2} \det(\Omega_t)^{-1/2} \exp\Big(-\tfrac{1}{2}\, a_t'\Omega_t^{-1} a_t \Big),$$

and taking logs gives (23), the exact Gaussian likelihood of the sample.

C Population Moments and Variance Decomposition

The stationary variance of the state solves $\Sigma = \mathcal{M}\Sigma\mathcal{M}' + \mathcal{C}\mathcal{C}'$. Since $s_{t+k} = \mathcal{M}^k s_t$ plus innovations independent of s_t, the autocovariances of the observations are $\Gamma_k \equiv \mathrm{Cov}(y_{t+k}, y_t) = \mathcal{G}\mathcal{M}^k\Sigma\mathcal{G}'$, from which the population moments in Table 2 follow. Let $\mathcal{C}_j$ be the jth column of $\mathcal{C}$. Because the components of ϵ_t are independent, $\mathcal{C}\mathcal{C}' = \sum_j \mathcal{C}_j\mathcal{C}_j'$, and by linearity of the Lyapunov equation $\Sigma = \sum_j \Sigma^{(j)}$ with $\Sigma^{(j)} = \mathcal{M}\Sigma^{(j)}\mathcal{M}' + \mathcal{C}_j\mathcal{C}_j'$. The share of shock j in the variance of series i is $[\mathcal{G}\Sigma^{(j)}\mathcal{G}']_{ii}/[\mathcal{G}\Sigma\mathcal{G}']_{ii}$; for the model variables in Table 3, the same computation uses M, C, and the two structural shocks only.